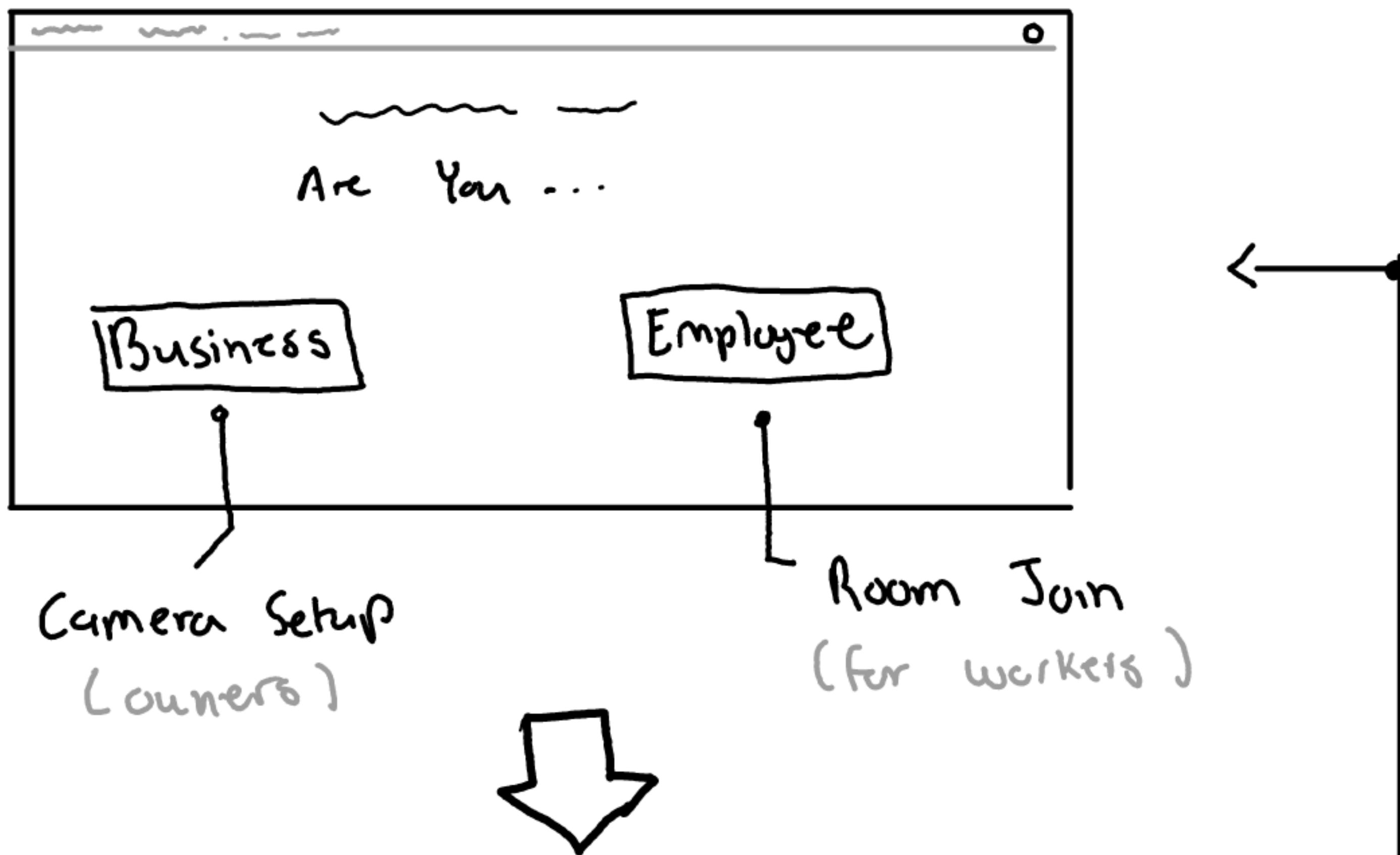
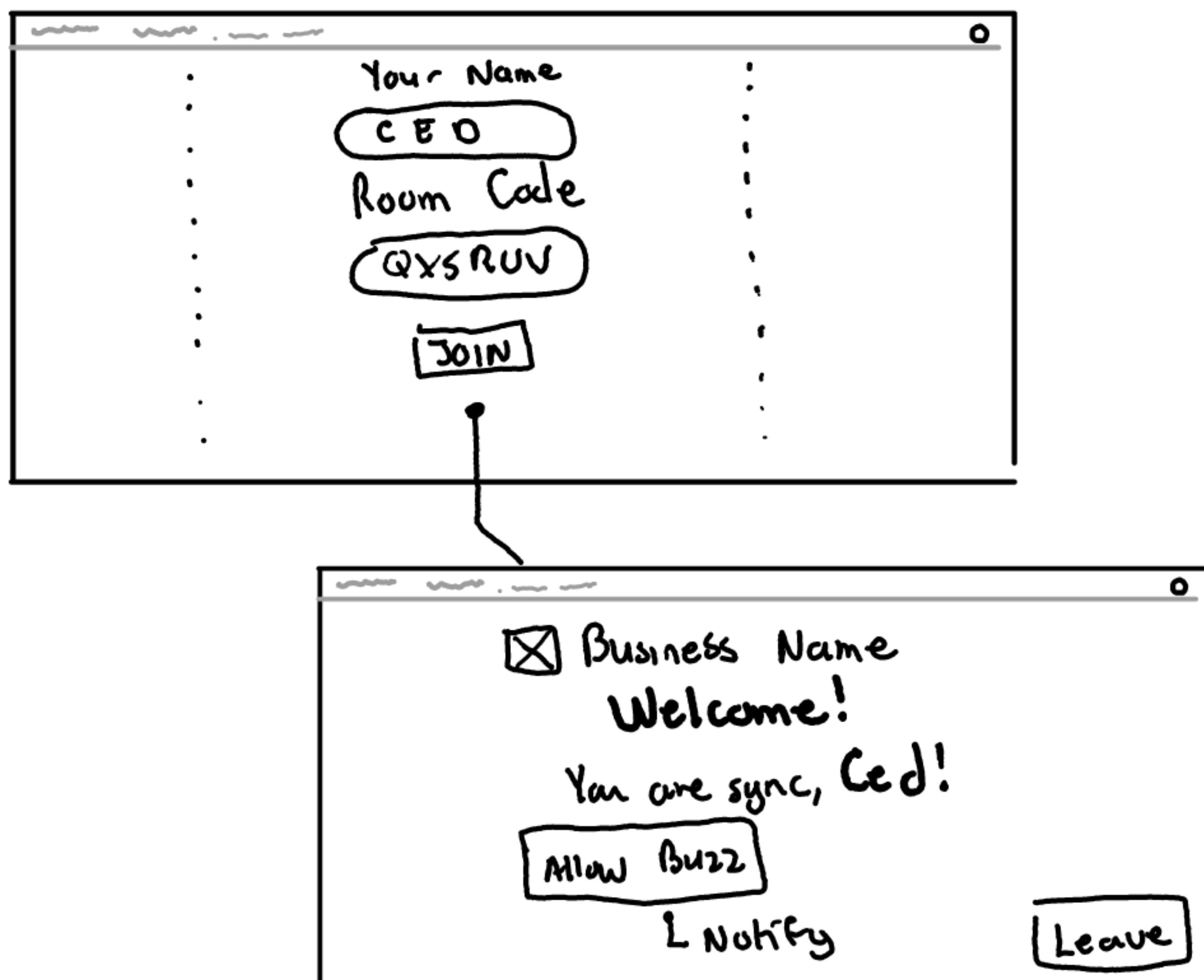


Restaurant Assistant

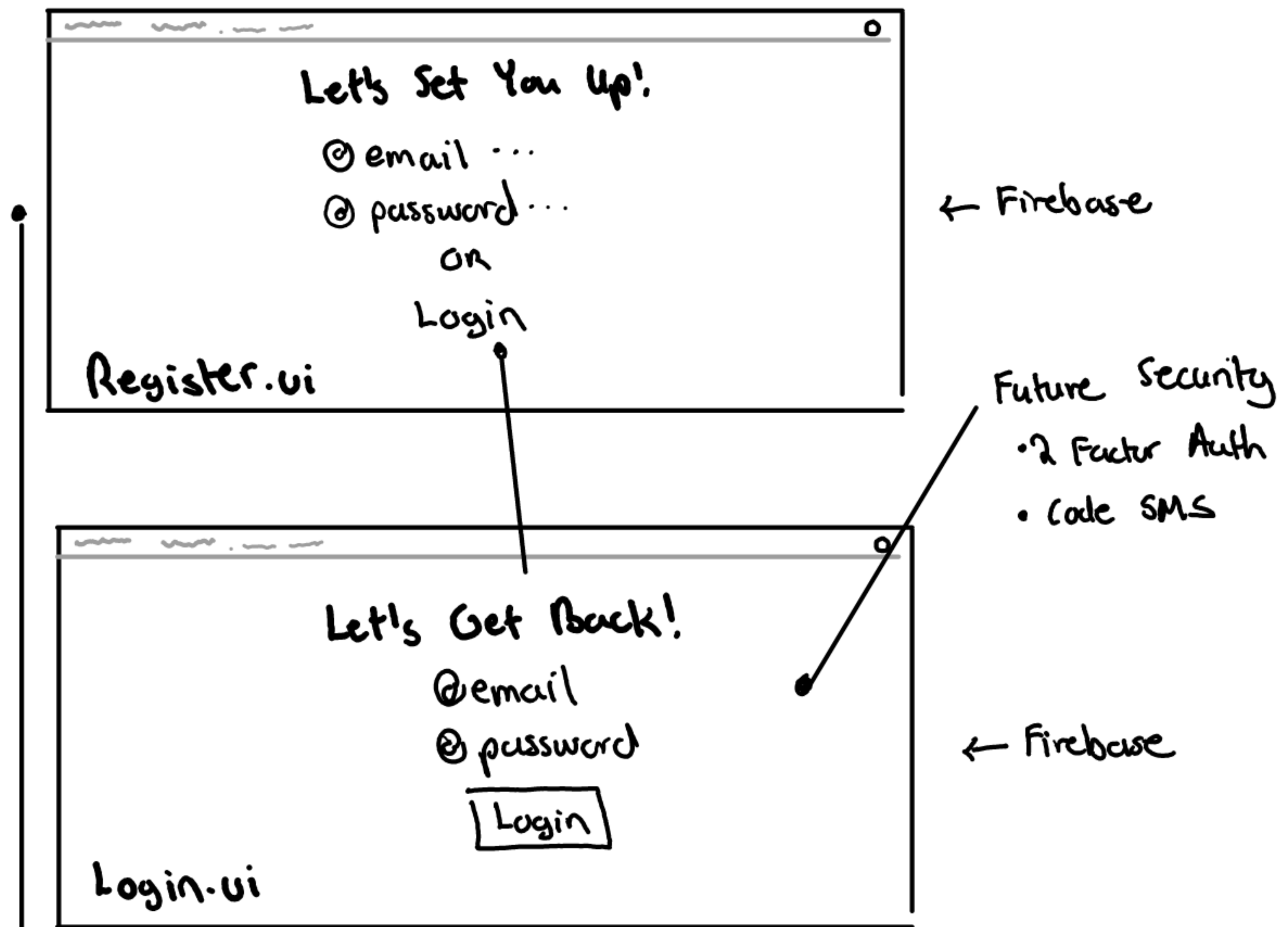
• Design Plan : UI Interfaces & Functionalities



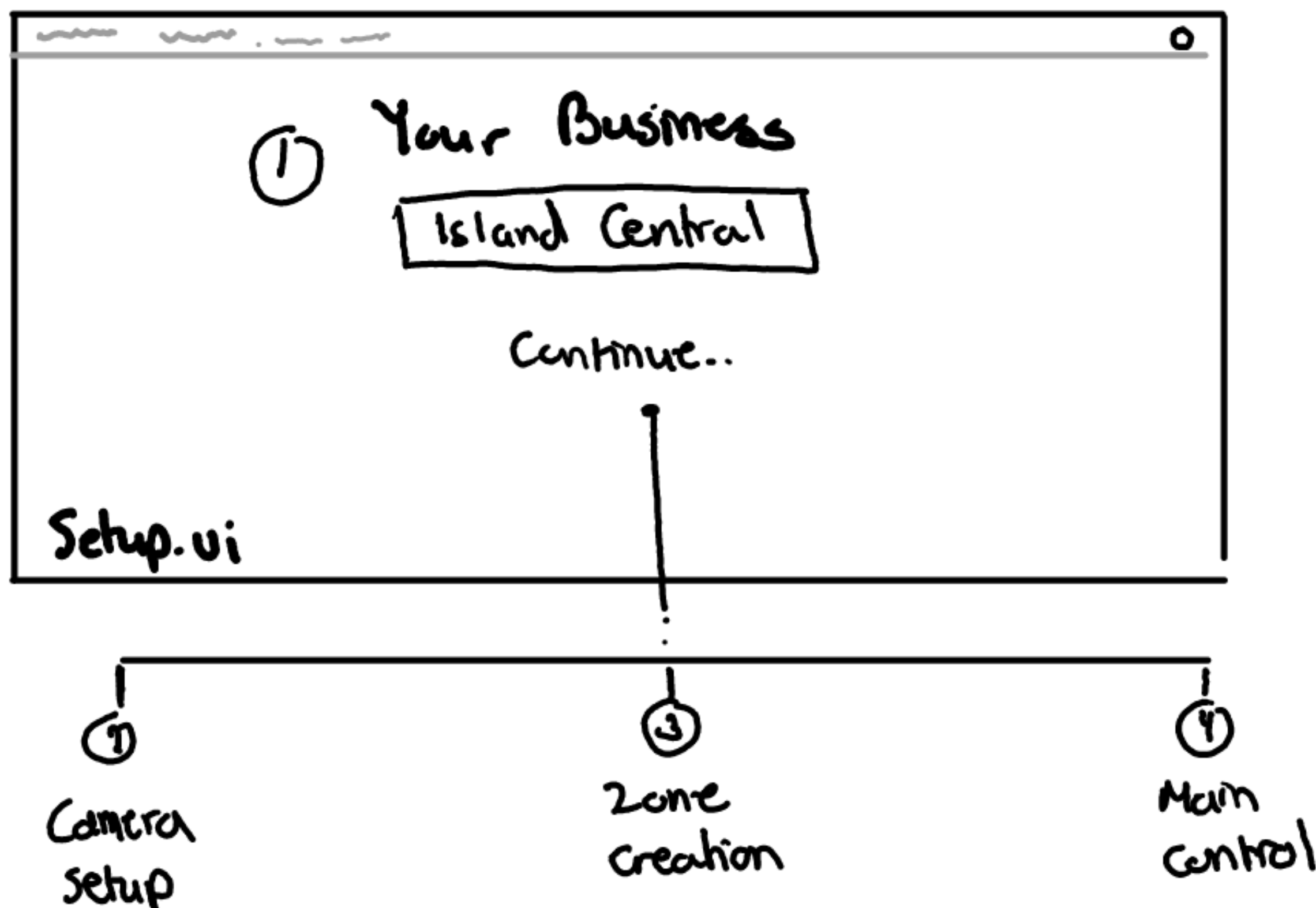
Room Joining (Employee UI)



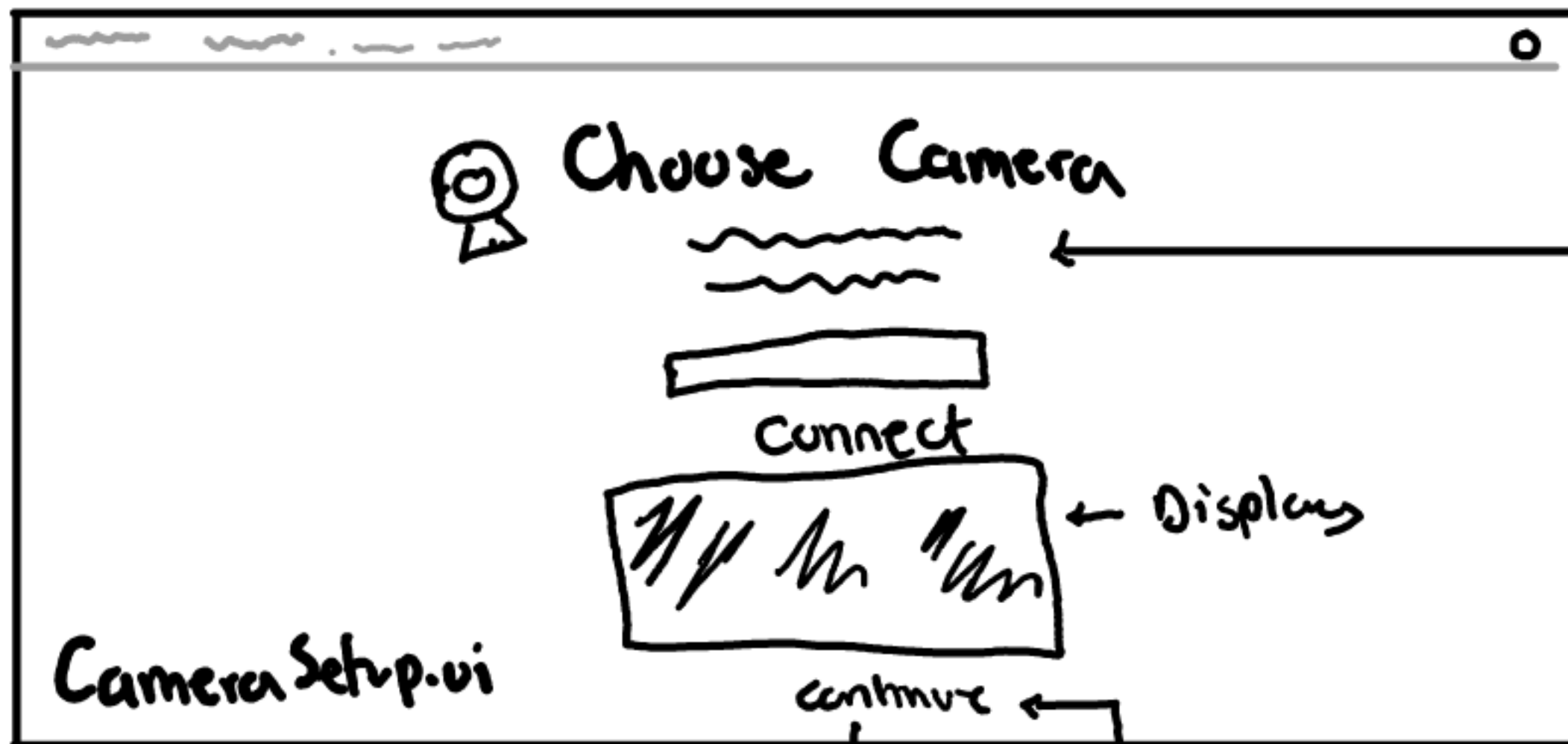
Business Owner Interface



Creation: If not setup yet (New)



Camera Setup



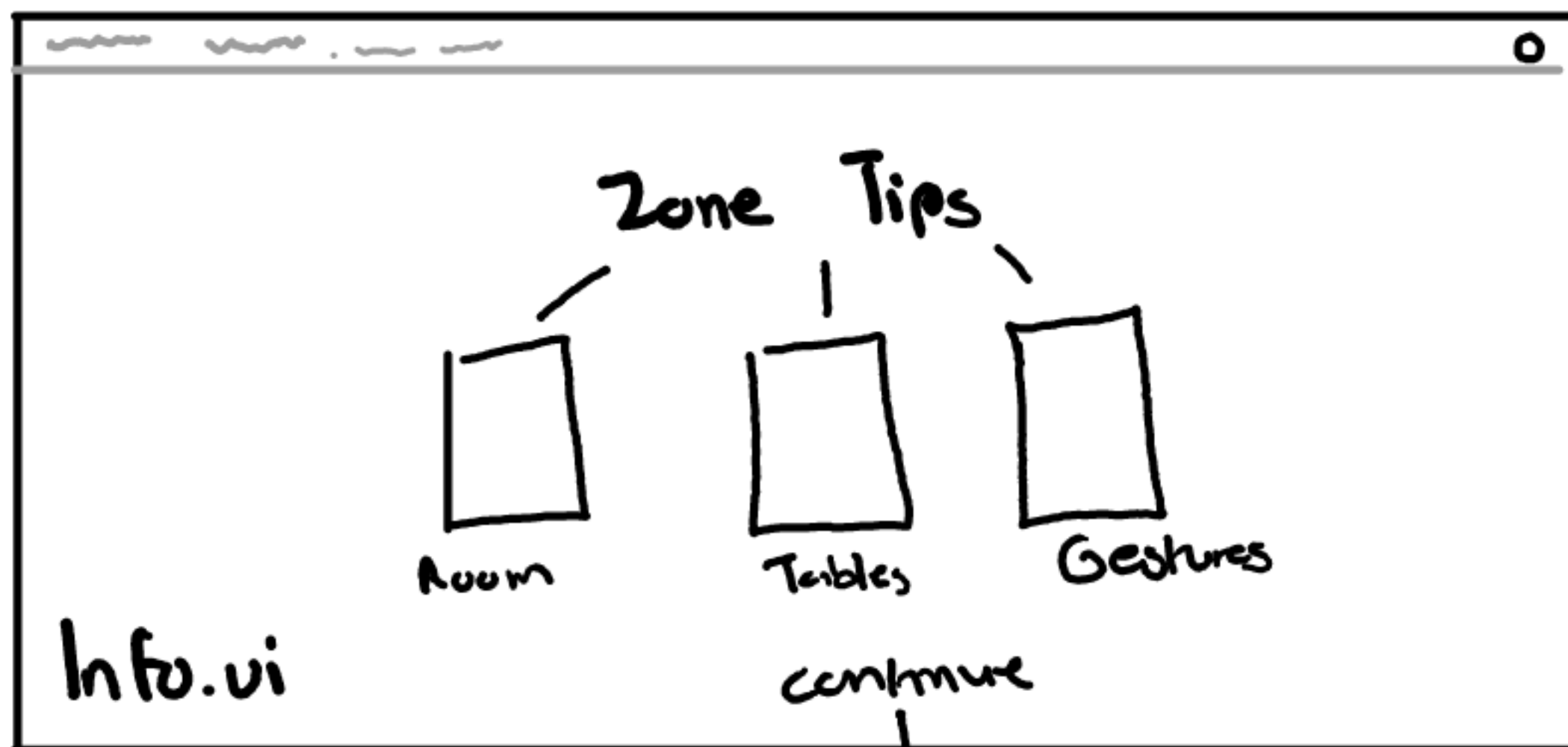
Suggestions on:

- Best Angle
- Placement
- Device Recs.

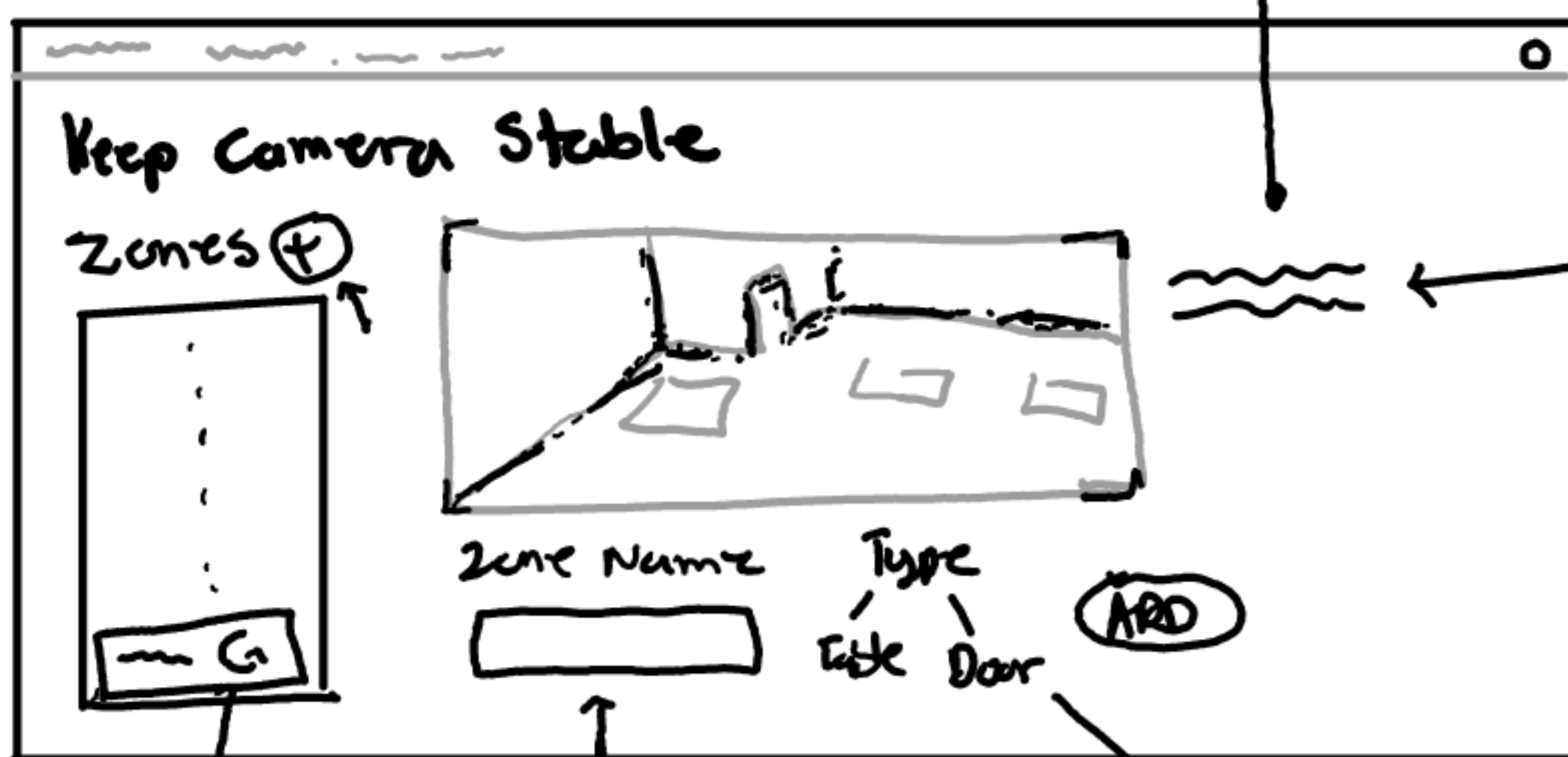
Zone Creation

Before Cont.

- Don't Move
- Ensure Location is good



Zone Manager.vi



users draw shape

Info / Tips

Data

- 2name
- box position
- 2type

edit zones

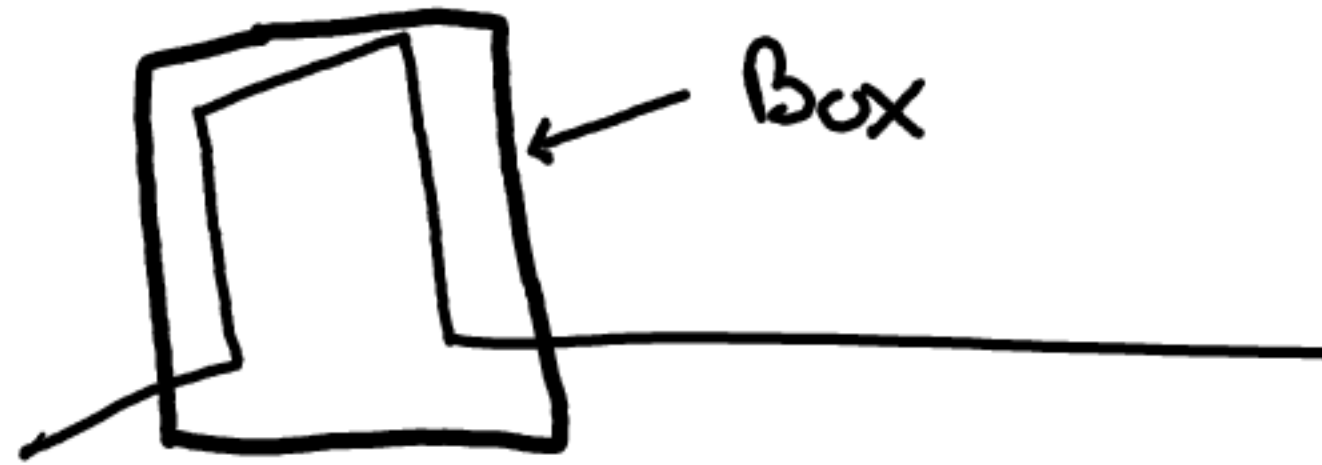
- Need:
- Door Zone
 - 1 table zone
- ↓
- then continue

Table or Door

Mandatory Zones

Door Zone

- customer entering detection
- Dasher detection



AI detects movement / face in this zone → Buzz Employees

Maybe handle zone creations in one view?

Table Zone

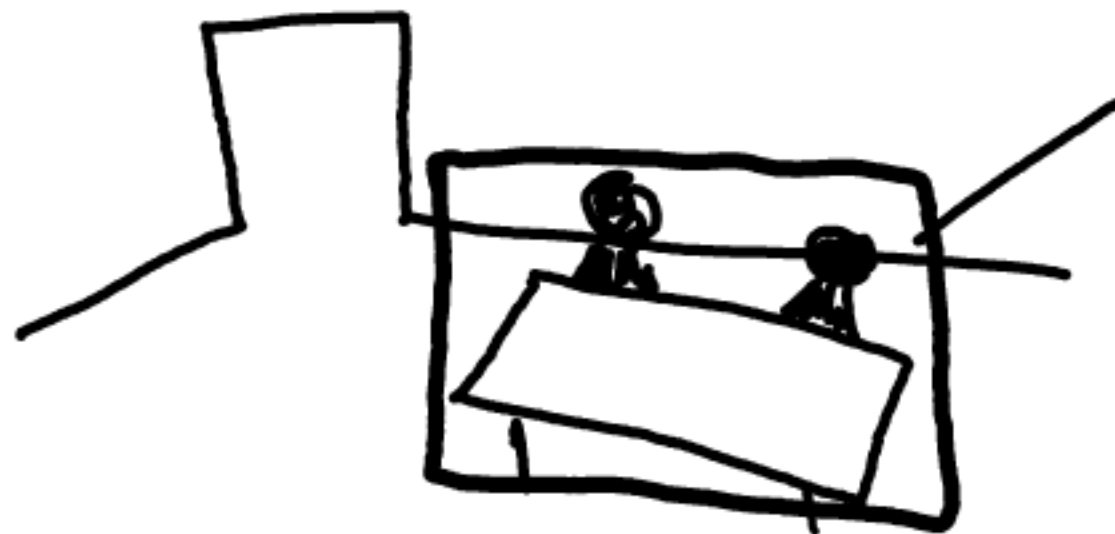
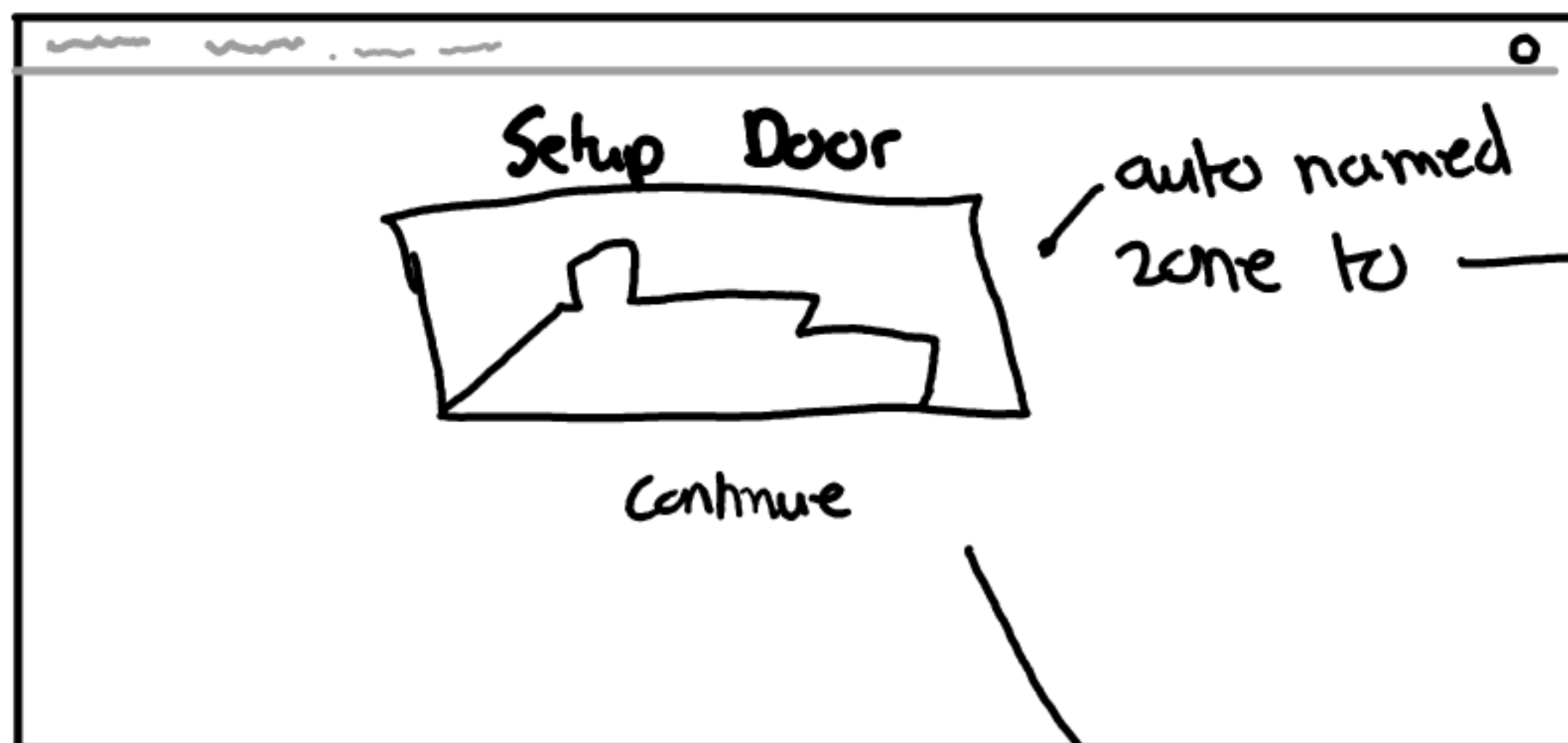


Table 1
• detects hand gestures
AI Trained Model
• gesture possibilities

Force user to setup Door + Table 1



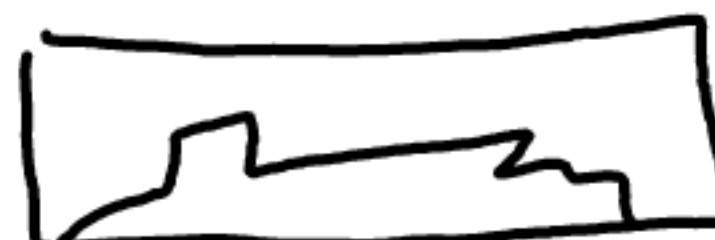
auto named zone to

Door Zone

Continue

"create more tables later"

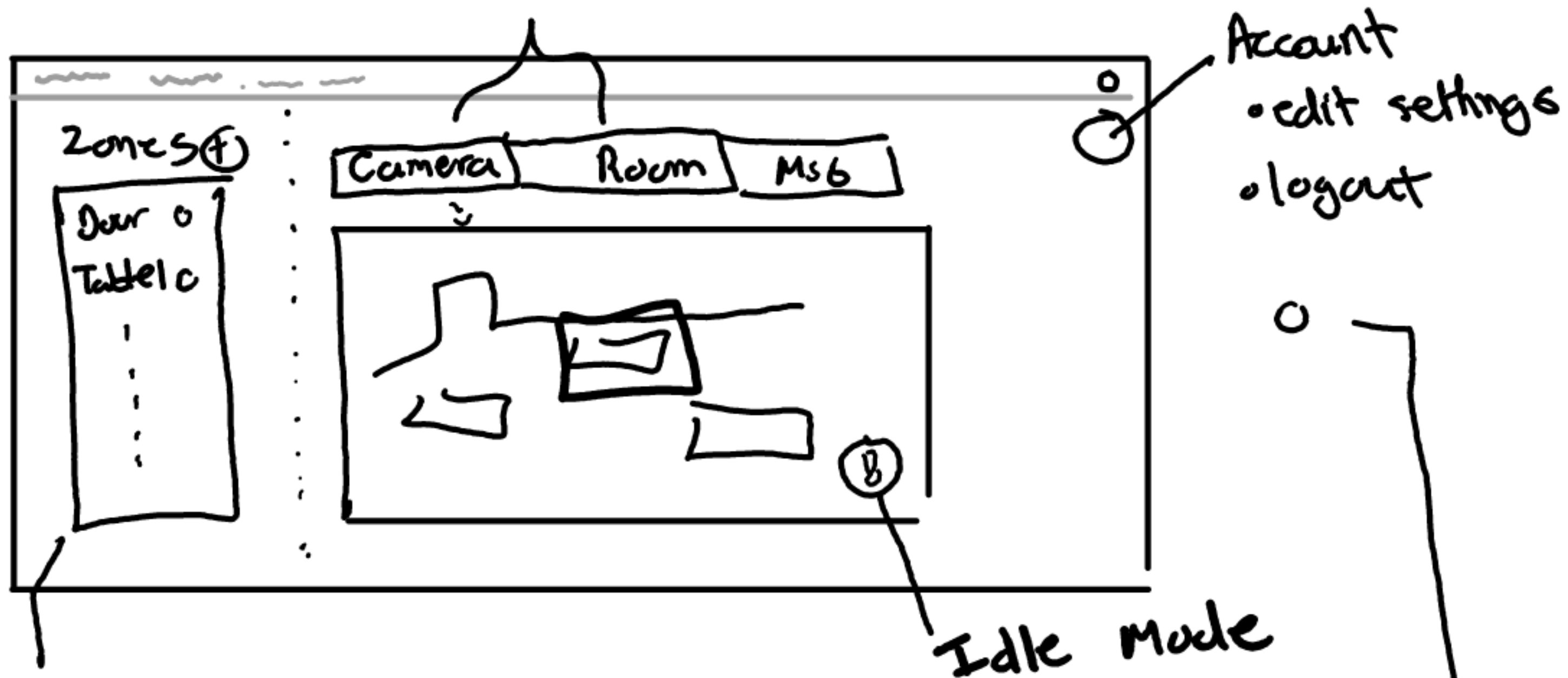
Setup Table 1



Continue → Main Hub.ui

MainHub.ui

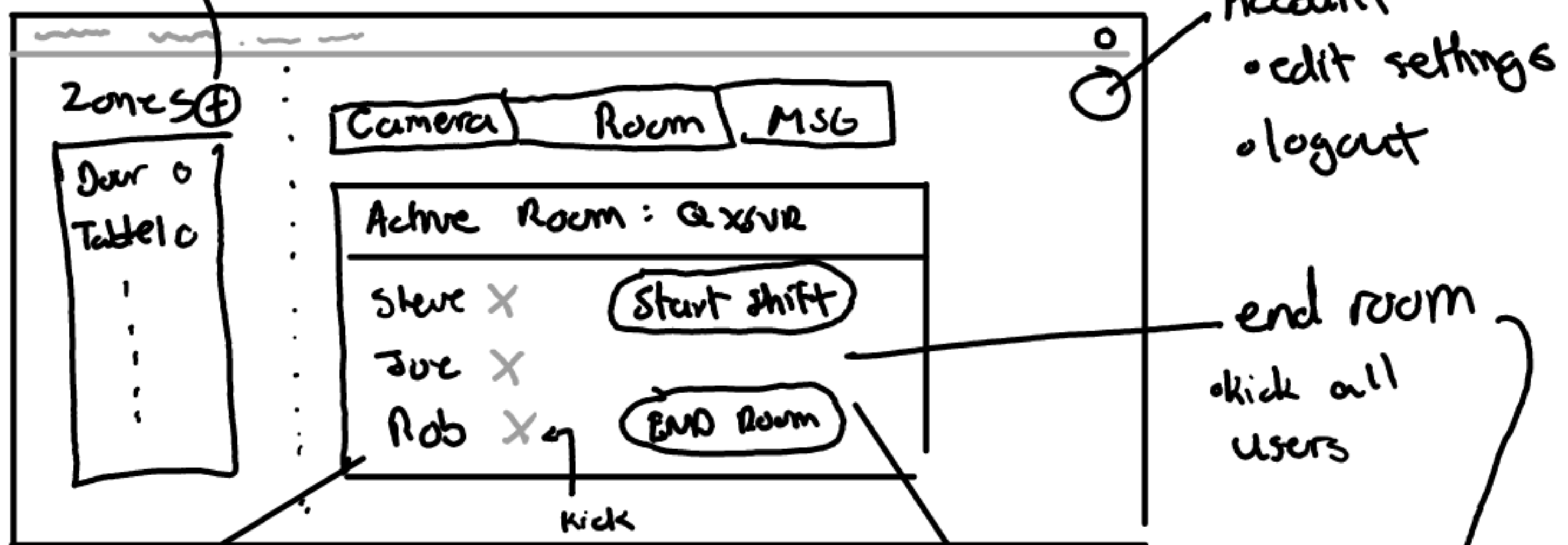
2 Buttons



Door & Table
• editable only
• can't be removed

Idle mode
Create Room
• Buzzer setup
for employees

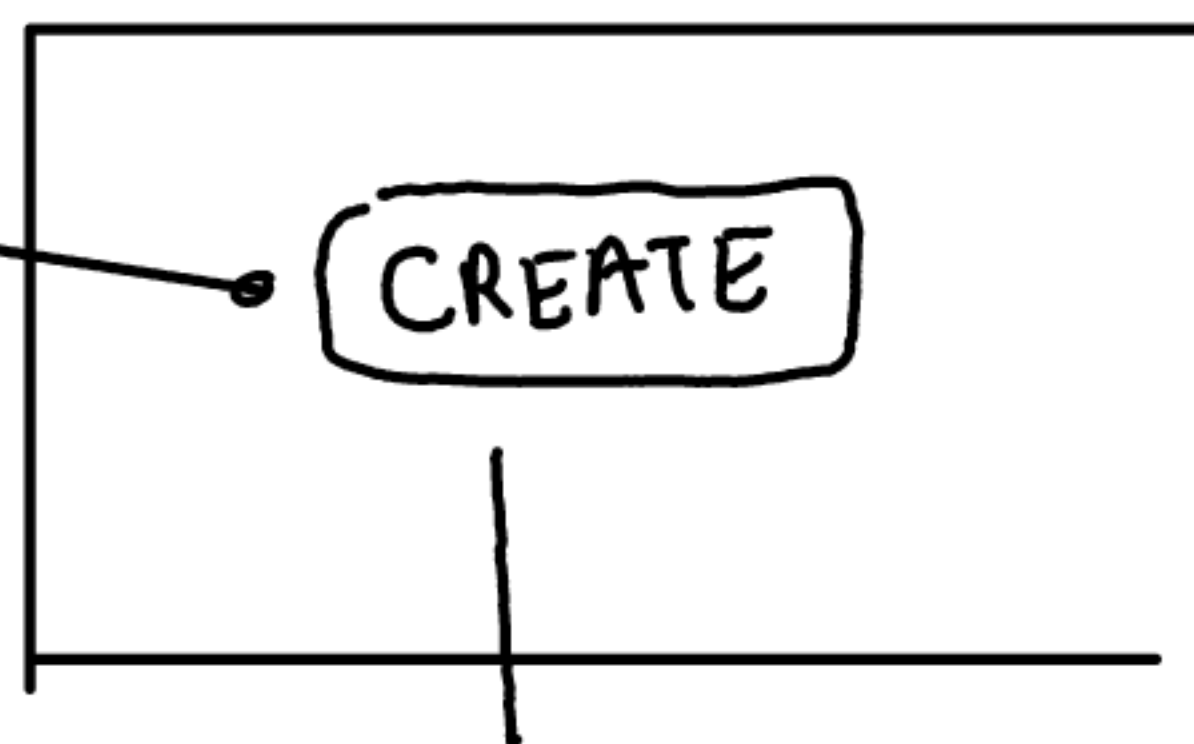
takes to ZoneManager.ui



Firebase: RoomId: Code: Active: memberlist
Generates
Code

Not active room

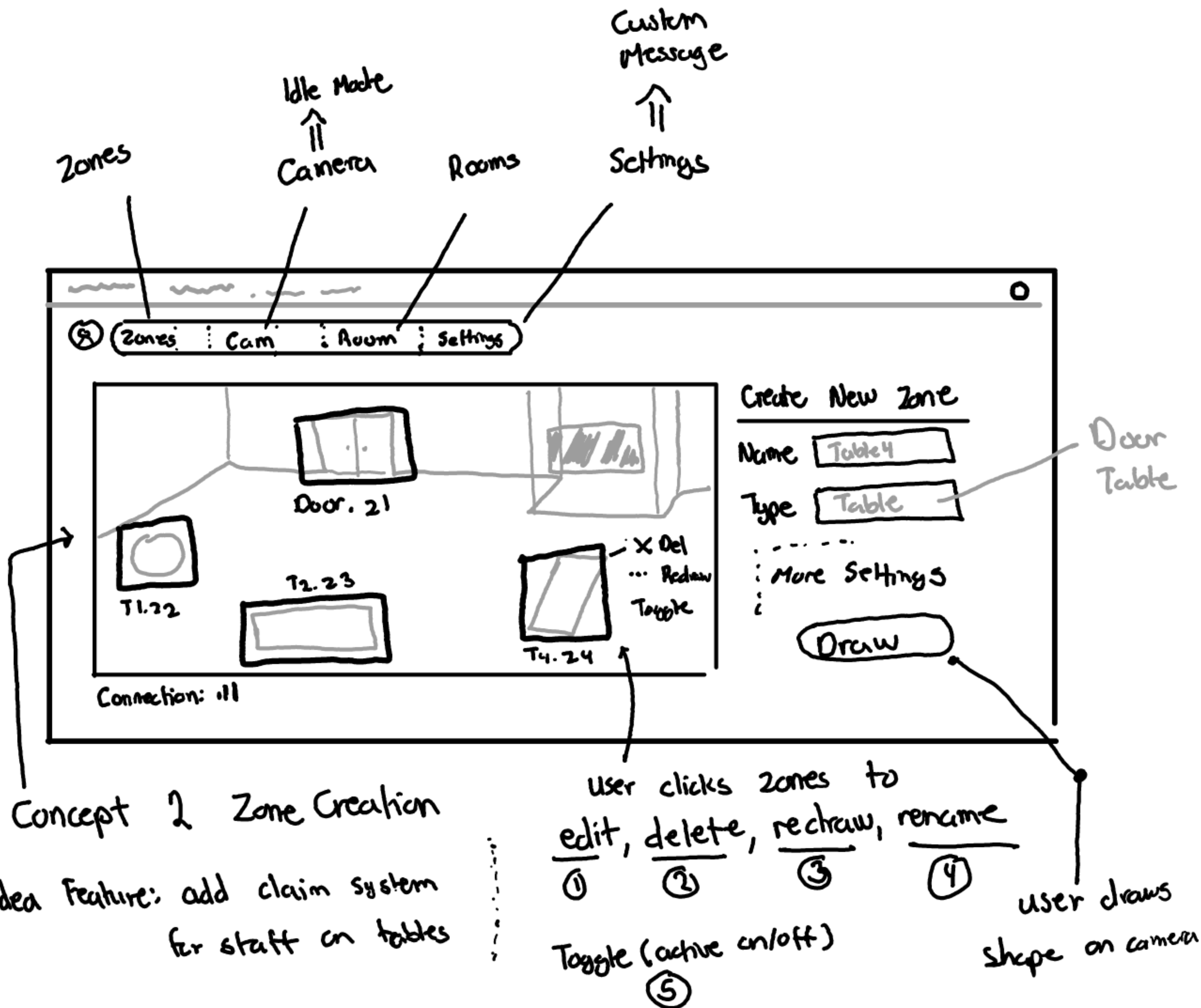
(rooms last only
for 24-hours)



Firebase: RoomId: Code: Active: Memberlist

Revised Zone Manager. ui

- centralized functionality



Firestore Data

name:

type:

enabled:

polygon: [{ 3]

created At:

version:

Account settings

Settings

email ~~~~~

Camera ~~~~~ change

Warning:

- must readjust all zones

Logout ←

changing camera position/device

- user needs to manually readjust the zone boxes

Logs user out

Zones ⊕

Camera	Room	MSG
EDIT MSG		
Door		"~~~~~"...
Table		"~~~~~"...

MSG for AI Detection

Camera > Idle Mode

- Idle Mode displays a giant flashing message to workers through device screen

Back

Customer Detected!

Zone: Door

camera

Idle Mode.ui Enable Disable

Door Specific MSG

Zone Name

- no msg
- no buzz

Key Data Store

- UserId: string
 - email: string
 - password: string
 - BusinessName: string ; BusinessId
-

name:

type:

enabled:

polygon: [{ 3]

created At:

version:

Zones

- Code: string
 - businessId: string
 - expires At: { duration }
 - active: boolean
 - created At: timestamp
-

Room
Code

- workerId: string
 - businessId: string
 - displayName: string
 - role: string
 - status: string (online, offline, removed)
 - joinedAt: timestamp
-

Room Scenarios

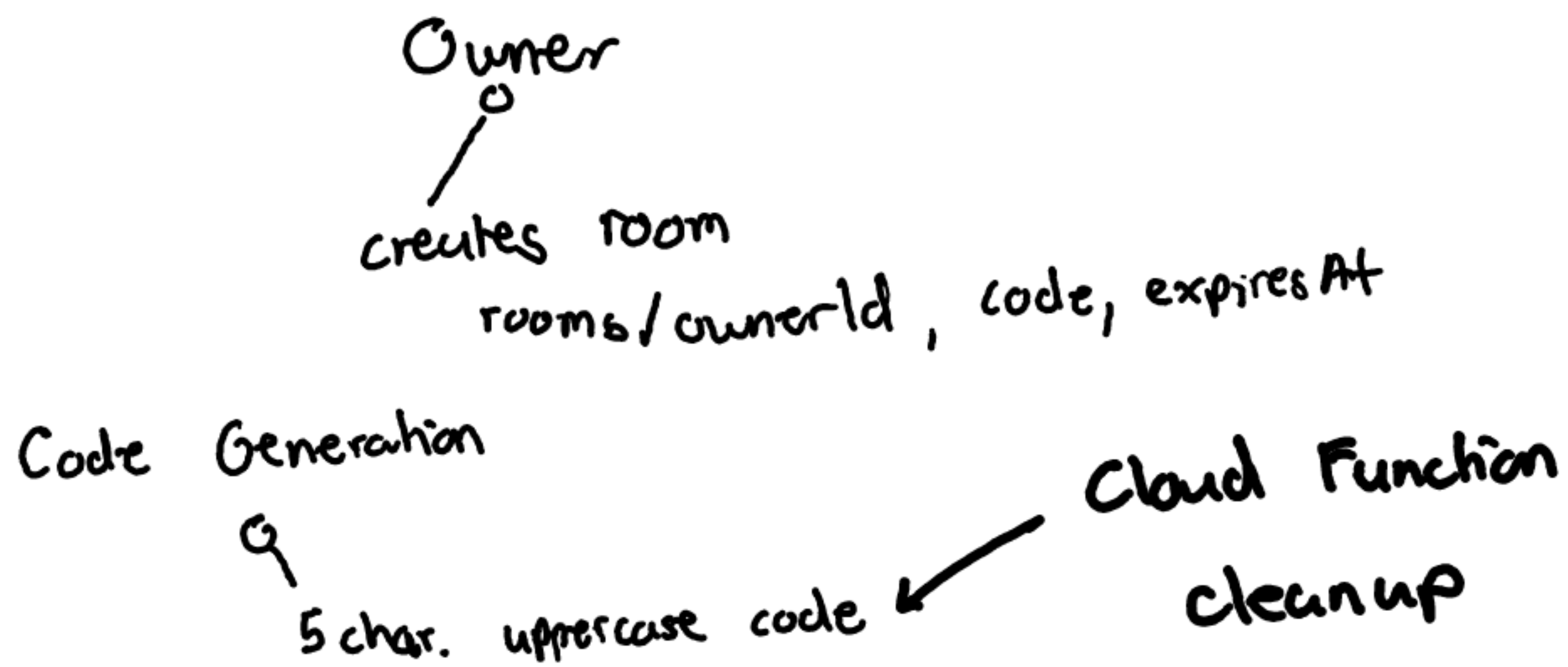
Don't Kick on

- invite expire
 - Owner generate new code
 - Camera changed / moved
-

Kick on

- owner kick
- leave room (by staff)
- room deleted
- shift ended

Firestore Room Generation



o Since workers can join via no Auth

⋮

Firebase Sign-In Methods

① email / password ← owners

② Anonymous ← workers

↳ change firestore rules as well

⋮

Firestore Database

rooms: collectionId

collection group

Expires At

← anonymous field

○ required rule adjustments